

Late Delivery Risk Prediction

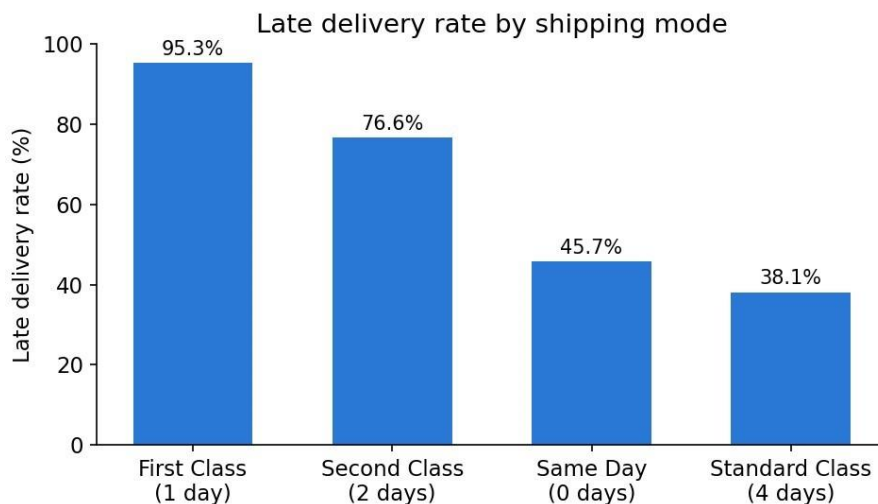
Executive Summary — APL Logistics | July 2026

APL Logistics can now flag, before an order ships, whether it's at risk of arriving late — with 82% accuracy at separating risky orders from safe ones, and a clean three-tier risk score (Low / Medium / High) that operations can act on directly. More importantly, the analysis identified the real driver of late deliveries, and it isn't what most teams assume.

172,762 orders analyzed	0.82 model accuracy (ROC-AUC)	88% actual late rate in "High Risk" tier	25% actual late rate in "Low Risk" tier
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The Core Finding

Late delivery risk is driven almost entirely by one thing: how aggressive the promised delivery window is for the shipping mode selected — not the region an order is going to, not the product category, not order size. Counterintuitively, faster promised shipping is far riskier than slower shipping, because tight windows are simply harder to hit.



First Class orders (1-day promise) are late 95% of the time; Standard Class (4-day promise) is late 38% of the time.

What This Means for the Business

- This is not primarily a fulfillment or carrier performance problem — it's a promise-setting problem. The delivery windows attached to expedited shipping options are structurally very hard to meet.
- Region, market, and product category — the factors ops teams often investigate first — showed almost no relationship to late delivery risk in this data.
- The risk score is well-calibrated: orders flagged "High Risk" are actually late 88% of the time, vs. 25% for "Low Risk" — reliable enough to drive real operational decisions.

Recommendations

- Use the risk score to proactively flag High Risk orders for priority handling or earlier carrier hand-off, before they ship — not after they're already late.
- Revisit the delivery-time promises on expedited shipping modes; either tighten fulfillment capacity to match the promise, or reset customer expectations to match reality.
- Invest in better data going forward — specifically order timestamps and order IDs — to unlock seasonal analysis and more precise risk modeling than is currently possible.

Deliverables

- Trained prediction model (XGBoost) with documented performance and explainability
- Interactive Streamlit dashboard for order-level risk lookup, region/mode analysis, and a daily operations action list
- Full research paper detailing methodology, findings, and limitations